

VELIKO TARNOVO, Bulgaria

WMO: 155300

Lat: 43.083N

Lon: 25.650E

Elev: 715

StdP: 14.32

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.5	15.8	4.4	7.1	11.3	10.1	9.5	17.2	30.8	45.2	22.8	43.5	1.3	270	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	94.9	70.1	91.3	69.3	88.3	68.6	73.4	87.6	71.7	85.8	70.1	84.1	5.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.0	109.3	80.1	67.2	102.6	77.6	65.7	97.4	75.9	37.4	87.8	35.9	85.5	34.5	84.4	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.6	15.3	11.8	DB	3.3	100.3	6.9	4.9	-1.7	103.8	-5.7	106.7	-9.6	109.4	-14.6	113.0	(4)
(5)			WB	2.3	76.7	6.8	3.6	-2.5	79.3	-6.5	81.4	-10.3	83.4	-15.2	86.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	33.5	37.2	45.0	54.2	63.4	70.1	74.2	74.0	65.3	55.2	45.2	35.4		
	DBStd	16.31	9.51	10.10	8.55	6.90	6.10	5.63	5.12	5.30	6.50	7.84	8.54	9.46		
	HDD50	1817	515	364	204	37	1	0	0	0	0	40	195	461		
	HDD65	4731	975	778	619	330	102	18	2	3	75	315	596	918		
	CDD50	3454	5	6	49	161	417	604	751	745	458	200	50	8		
	CDD65	894	0	0	0	4	54	172	288	282	83	10	1	0		
	CDH74	9641	0	0	7	64	633	1759	3116	3052	871	134	5	0		
	CDH80	3934	0	0	0	11	143	654	1412	1401	286	27	0	0		
(14) Wind	WSAvg	3.3	3.2	3.7	4.4	4.1	3.4	3.3	3.2	2.9	3.0	2.6	3.1	2.8		
(15) Precipitation	PrecAvg	25.6	1.7	1.7	1.9	2.1	3.1	2.7	2.6	1.9	2.5	1.8	1.6	2.2		
	PrecMax	44.3	4.0	3.5	6.0	4.4	6.6	5.4	8.0	5.5	8.7	4.5	3.8	3.8		
	PrecMin	16.7	0.4	0.2	0.1	0.1	0.9	0.4	0.2	0.2	0.1	0.0	0.1	0.3		
	PrecStd	6.4	0.8	0.9	1.3	1.1	1.7	1.6	2.1	1.4	2.0	1.1	1.0	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.8	66.5	74.2	81.3	88.0	94.3	101.0	99.6	91.9	83.7	73.7	64.6		
		MCWB	51.1	51.7	57.0	61.2	67.0	69.8	70.0	69.6	67.0	65.5	60.0	53.9		
	2%	DB	57.3	61.4	68.6	74.7	83.8	90.3	95.5	94.9	86.9	77.7	68.3	59.0		
		MCWB	49.1	49.9	53.6	57.9	64.3	70.4	70.1	70.4	65.4	62.7	58.6	50.9		
	5%	DB	53.0	57.4	64.7	70.8	80.6	86.8	91.3	91.2	83.0	73.1	64.1	54.1		
		MCWB	46.1	48.0	51.3	56.2	63.4	69.3	69.4	69.1	64.8	61.2	55.8	48.3		
	10%	DB	47.7	53.0	60.3	67.6	77.4	83.6	87.7	88.1	78.9	68.7	59.4	49.6		
		MCWB	42.7	45.4	49.3	54.5	62.4	67.8	68.9	68.2	63.3	58.9	52.9	45.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	53.0	57.9	62.2	69.1	74.6	75.6	75.7	69.1	66.6	62.4	54.5		
		MCDB	60.1	64.4	70.9	77.3	82.9	87.0	91.0	88.7	81.5	80.3	69.3	62.6		
	2%	WB	49.5	50.6	54.4	59.2	66.7	72.7	72.9	73.4	67.4	64.1	58.6	51.2		
		MCDB	56.7	60.1	66.7	71.0	79.8	86.2	87.8	87.4	82.3	75.5	67.5	57.5		
	5%	WB	46.5	48.3	52.2	57.4	65.1	70.7	71.4	71.2	65.9	61.7	56.1	48.6		
		MCDB	52.6	56.9	63.2	67.9	77.2	84.1	85.5	85.7	80.1	71.1	63.3	54.0		
	10%	WB	42.8	45.7	49.8	55.8	63.4	68.9	70.0	69.7	64.5	59.5	53.2	45.3		
		MCDB	47.4	52.7	59.4	65.4	74.5	81.2	83.8	84.7	76.6	67.2	58.8	49.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	17.1	17.8	19.1	20.9	21.0	22.1	22.9	20.5	18.6	15.8	13.8		
		MCDBR	23.1	26.1	26.8	25.6	26.8	27.1	29.1	29.2	28.0	26.0	23.8	21.0		
		MCWBR	14.8	15.8	13.9	11.9	11.3	10.3	9.8	9.3	10.5	13.0	13.9	14.0		
	5% WB	MCDBR	22.0	24.6	24.7	21.8	23.0	24.4	24.0	24.6	24.4	23.7	20.8	18.6		
		MCWBR	14.7	15.2	13.1	11.1	10.6	10.2	9.9	9.6	10.1	12.3	12.8	12.9		
(40) Clear-Sky Solar Irradiance	taub		0.345	0.378	0.423	0.481	0.456	0.461	0.468	0.481	0.447	0.405	0.371	0.347		
	taud		2.281	2.201	2.121	1.973	2.118	2.146	2.137	2.096	2.153	2.251	2.299	2.286		
	Ebn at Noon		243	253	256	250	261	259	256	247	246	241	231	229		
	Edh at Noon		28	36	44	55	49	48	48	48	42	34	28	26		
(44) All-Sky Solar Radiation	RadAvg		485	736	1080	1482	1836	2009	2056	1850	1356	882	529	403		
	RadStd		69	94	97	158	147	147	148	111	105	95	66	54		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	+0.79	N/A	N/A	N/A	+0.81	N/A	N/A	-234	+130	N/A

Nomenclature: See separate page